Dongsheng Ding

CONTACT INFORMATION

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville
1520 Middle Drive, Knoxville, TN 37996

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TEACHING INTERESTS

I view teaching as a responsibiity toward future generations. Technology rapidly advances, yet engineering course materials stay relatively static, limiting alignment with today's interdisciplinary innovations. My teaching aims to develop forward-thinking curricula that bridge interdisciplinary gaps and expand career possibilities.

RESEARCH INTERESTS

Application of principles and tools in optimization and control to the *study of requirement-driven machine decision-making*.

Topics of interest span: constrained reinforcement learning; multi-agent reinforcement learning; constrained generative modeling; constrained alignment of diffusion and large language models; robustness and resilience analysis of optimization algorithms; nonlinear control of fractional-order dynamical systems.

Applications of interest include: autonomous robots/vehicles; generative artificial intelligence in applied science and engineering, such as natural language processing, image processing, and molecular design; decision-making systems for power grid, transportation, communication, and healthcare management.

APPOINTMENT

Assistant Professor — August 2025 – Now
University of Tennessee, Knoxville

Postdoctoral Researcher — September 2022 – July 2025
University of Pennsylvania

Graduate Research Assistant — August 2017 – August 2022
University of Southern California

Graduate Research Assistant & Teaching Assistant — August 2015 – August 2017
University of Minnesota, Twin Cities

EDUCATION

PhD in Electrical Engineering, GPA: 4.0/4.0 — August 2022
University of Southern California
Thesis: Provable reinforcement learning for constrained and multi-agent control systems
Advisor: Mihailo R. Jovanović

MS in Electrical Engineering, GPA: 3.9/4.0 — August 2017
University of Minnesota, Twin Cities

ME in Control Theory & Engineering, GPA: 3.7/4.0 — March 2015
BE in Automation, GPA: 3.8/4.0 — June 2011
Zhejiang University, Hangzhou, China

SUBMITTED PAPERS

1. T. Wang, S. Li, Y. Sun, D. Ding, E. Dobriban. "Where to spend rollouts: Hit-utility optimal rollout allocation for group-based RLVR," 2026; also arXiv: 2605.07114. (under review)
2. S. Khalafi*, K. Fallah*, S. Cho, D. Ding, J. Anderson, A. Ribeiro. "ConicFlow: Constrained Inference-time optimal control for flow matching models," 2026; also arXiv: 2610.14307. (under review, *equal contribution)

REFEREED PUBLICATIONS

Journals

1. D. Ding, K. Zhang, J. Duan, T. Başar, and M. R. Jovanović. "Convergence and sample complexity of natural policy gradient primal-dual methods for constrained MDPs," *J. Mach. Learn. Res.*, 26(256) 1-76, 2025; also arXiv: 2206.02346.
2. Q. Wang, J. Zhang, D. Ding, and D. Qi, "Adaptive Mittag-Leffler stabilization of a class of fractional order uncertain nonlinear systems," *Asian J. Control*, 18(6) 2343–2351, 2016.
3. D. Ding, D. Qi, and Q. Wang, "Asymptotic pseudo-state stabilization of uncertain fractional-order nonlinear systems with additive disturbance," *Nonlinear Dyn.*, 81(1) 667–677, 2015.
4. Q. Wang, D. Ding, and D. Qi, "Mittag-Leffler synchronization of uncertain fractional order chaotic systems," *Chinese Physics B*, 24(6), 2015.
5. D. Ding, D. Qi, and Q. Wang, "Nonlinear Mittag-Leffler stabilization of commensurate fractional-order nonlinear systems," *IET Control Theory Appl.*, 9(5) 681–690, 2014.
6. D. Ding, D. Qi, X. Luo, J. Chen, X. Wang, and P. Du, "Convergence analysis and performance of an extended central force optimization algorithm," *Appl. Math. Comput.*, 219(4), 2246–2259, 2012.
7. D. Ding, X. Luo, J. Chen, X. Wang, P. Du, and Y. Guo, "A convergence proof and parameter analysis of central force optimization algorithm," *J. Convergence Inf. Technol.*, 6(10), 16–23, 2011.

Machine Learning and Artificial Intelligence Conferences

1. S. Khalafi, A. Ribeiro, and D. Ding*. "Unlearning in diffusion models: A unified framework with KL divergence and likelihood constraints," in *Proceedings of the International Conference on Machine Learning*, Seoul, South Korea, 2026; also arXiv:2605.30825. (**acceptance rate** 26.6%, ***corresponding author**)
2. B. Zhang, S. Li, I. Hounie, O. Bastani, D. Ding*, A. Ribeiro. "Alignment of large language models with constrained learning," in *Proceedings of the Advances in Neural Information Processing Systems*, San Diego, California, 2025; also arXiv:2505.19387. (**acceptance rate** 24.5%, ***corresponding author**)
3. S. Khalafi, I. Hounie, D. Ding*, A. Ribeiro. "Composition and alignment of diffusion models using constrained learning," in *Proceedings of the Advances in Neural*

Information Processing Systems, San Diego, California, 2025; also arXiv:2508.19104. (**acceptance rate** 24.5%, ***corresponding author**)

4. S. Rozada, D. Ding*, A. Marques, A. Ribeiro. "Deterministic policy gradient primal-dual methods for continuous-space constrained MDPs." in *Proceedings of the AAAI Conference on Artificial Intelligence*, Philadelphia, Pennsylvania, 2025; also arXiv:2408.10015. (**acceptance rate** 23.4%, ***corresponding author**)
5. X. Huang, S. Li, E. Dobriban, O. Bastani, H. Hassani, and D. Ding*. "One-shot safety alignment for large language models via optimal dualization." in *Proceedings of the Advances in Neural Information Processing Systems*, Vancouver, Canada, 2024; also arXiv: 2405.19544. (**acceptance rate** 25.8%**, spotlight**, ***corresponding author**)
6. S. Khalafi, D. Ding*, and A. Ribeiro. "Constrained diffusion models via dual training." in *Proceedings of the Advances in Neural Information Processing Systems*, Vancouver, Canada, 2024; also arXiv:2408.15094. (**acceptance rate** 25.8%, ***corresponding author**)
7. D. Ding, Z. Huan, and A. Ribeiro. "Resilient constrained reinforcement learning." in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, Valencia, Spain, 2024; also arXiv:2312.17194. (**acceptance rate** 27.6%)
8. D. Ding, C.-Y. Wei, K. Zhang, and A. Ribeiro. "Last-iterate convergent policy gradient primal-dual methods for constrained MDPs," in *Proceedings of the Advances in Neural Information Processing Systems*, New Orleans, Louisiana, 2023; also arXiv:2306.11700. (**acceptance rate** 26.1%)
9. D. Ding, X. Wei, Z. Yang, Z. Wang, and M. R. Jovanović. "Sample efficient genearalized Lagrangian policy optimization for safe multi-agent reinforcement learning," in *Proceedings of the Learning for Dynamics and Control Conference*, Philadelphia, Pennsylvania, 2023; also arXiv:2306.00212.
10. D. Ding, C.-Y. Wei, K. Zhang, and M. R. Jovanović. "Independent policy gradient for large-scale Markov potential games: sharper rates, function approximation, and game-agnostic convergence," in *Proceedings of the 39th International Conference on Machine Learning*, Baltimore, Maryland, 2022; also arXiv:2202.04129. (**acceptance rate** 21.5%**, 118/1117 long presentations**)
11. D. Ding, X. Wei, Z. Yang, Z. Wang, and M. R. Jovanović. "Provably efficient safe exploration via primal-dual policy optimization," in *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, Virtual, 2021; also arXiv:2003.00534. (**acceptance rate** 30%**, 48/455 orals**)
12. D. Ding, K. Zhang, T. Başar, and M. R. Jovanović. "Natural policy gradient primal-dual method for constrained Markov decision processes," in *Proceedings of the Advances in Neural Information Processing Systems*, Virtual, 2020. (**acceptance rate** 20%)
13. D. Ding, X. Wei, Z. Yang, Z. Wang, and M. R. Jovanović. "Fast multi-agent temporal-difference learning via homotopy stochastic primal-dual method," in *the*

Optimization Foundations for Reinforcement Learning Workshop at NeurIPS, Vancouver, Canada, 2019; also arXiv:1908.02805.

Control Conferences

1. D. Ding. "Convergence of natural policy gradient primal-dual methods for constrained convex MDPs," in *Proceedings of the 65th IEEE Conference on Decision and Control*, Honolulu, Hawaii, 2026. (accepted and to appear)
2. D. Ding and M. R. Jovanović. "Policy gradient primal-dual mirror descent for constrained MDPs with large state spaces," in *Proceedings of the 61st IEEE Conference on Decision and Control*, Cancún, Mexico, 2022.
3. D. Ding, K. Zhang, T. Basar and M. R. Jovanović, "Convergence and optimality of policy gradient primal-dual method for constrained Markov decision processes," in *Proceedings of the 2022 American Control Conference*, Atlanta, Georgia, 2022.
4. D. Ding, X. Wei, H. Yu, and M. R. Jovanović. "Byzantine-resilient distributed learning under constraints," in *Proceedings of the 2021 American Control Conference*, Virtual, 2021.
5. D. Ding, J. Yuan, and M. R. Jovanović. "Discounted online Newton method for time-varying time series prediction," in *Proceedings of the 2021 American Control Conference*, Virtual, 2021.
6. D. Ding and M. R. Jovanović. "Global exponential stability of primal-dual gradient flow dynamics based on the proximal augmented Lagrangian: A Lyapunov-based approach," in *Proceedings of the 59th IEEE Conference on Decision and Control*, Virtual, 2020.
7. D. Ding, X. Wei, and M. R. Jovanović. "Distributed robust statistical learning: Byzantine mirror descent," in *Proceedings of the 58th IEEE Conference on Decision and Control*, Nice, France, 2019.
8. D. Ding and M. R. Jovanović. "Global exponential stability of primal-dual gradient flow dynamics based on the proximal augmented Lagrangian," in *Proceedings of the 2019 American Control Conference*, Philadelphia, Pennsylvania, 2019.
9. D. Ding, B. Hu, N. K. Dhingra, and M. R. Jovanović. "An exponentially convergent primal-dual algorithm for nonsmooth composite minimization," in *Proceedings of the 57th IEEE Conference on Decision and Control*, Miami Beach, Florida, 2018.
10. D. Ding and M. R. Jovanović. "A primal-dual Laplacian gradient flow dynamics for distributed resource allocation problems," in *Proceedings of the 2018 American Control Conference*, Milwaukee, Wisconsin, 2018.
11. D. Ding, D. Qi, and Q. Wang, "Adaptive Mittag-Leffler stabilization of commensurate fractional-order nonlinear systems," in *Proceedings of the 53rd IEEE Conference on Decision and Control*, Los Angeles, California, 2014.

12. D. Ding, G. Zhang, D. Qi, and H. Zhang, "Strategy analysis of an evolutionary spectrum sensing game," in *Proceedings of the Intelligent Computing and Applications (LSMS & ICSEE)*, Shanghai, China, 2014. (**nominate paper award**)
13. D. Ding, D. Qi, and Q. Wang, "Alternative LMI characterizations for fractional-order linear systems," in *Proceedings of the 33rd Chinese Control Conference*, Nanjing, China, 2014.
14. D. Ding, D. Qi, and Q. Wang, "Fractional-order integral state space modeling and quasi state analysis via block operational matrix scheme," in *Proceedings of the 26th Chinese Control and Decision Conference*, Changsha, China, 2014.

RESEARCH PRESENTATIONS

Invited Talks & Posters

1. Invited talk of "Controllable generative diffusion models: A constrained optimization view," in *the INFORMS 2026 Annual Meeting*, San Francisco, California, 2026.
2. Invited talk of "Constrained policy optimization: A tale of regularization and optimism," in *the IFAC Workshop on Contextual Control: Integrating Learning, Optimization, and Physical Structures in Control Systems*, Busan, Republic of Korea, 2026.
3. Invited talk of "Controllable generative diffusion models: A constrained optimization view," in *the Yusu Wang's Group Seminar*, University of California, San Diego, California, 2026.
4. Invited talk of "Constrained policy optimization for large language model alignment," in *the Statistics and Data Science Seminar*, University of Tennessee, Knoxville, 2025.
5. Invited talk of "Constrained policy optimization for large language model alignment," in *the SIAM NNP Section Conference*, State College, Pennsylvania, 2025.
6. Invited talk of "Requirement-driven machine decision-making: From reinforcement learning to generative artificial intelligence," in *the EECS Faculty Candidate Seminar*, University of Tennessee, Knoxville, 2025.
7. Invited talk of "Multi-agent reinforcement learning for large-scale Markov potential games," in *the Workshop on Multi-Agent Learning in Dynamic Environments*, Texas A&M Institute of Data Science, TAMU, 2024.
8. Invited talk of "Multi-agent reinforcement learning for large-scale Markov potential games," in *the ESE PhD Colloquium*, UPenn, 2024.
9. Invited talk of "Constrained policy optimization: A tale of regularization and optimism," in *the INFORMS Optimization Society Conference*, Houston, Texas, 2024.
10. Invited talk of "Constrained policy optimization: A tale of regularization and optimism," in *the ESE PhD Colloquium*, UPenn, 2023.

11. Invited talk of "Provable constrained policy optimization in reinforcement learning," in *the Safe Reinforcement Learning Online Seminar*, Virtual, 2023.
12. Invited talk of "Finite-time performance of policy optimization methods for constrained reinforcement learning," in *the INFORMS 2022 Annual Meeting*, Indianapolis, Indiana, 2022.
13. Invited poster of "Independent policy gradient for large-scale Markov potential games" in *the REAL@USC-Meta center workshop*, ECE, USC, 2022.
14. Invited poster of "Provably efficient safe exploration via primal-dual policy optimization" in *the 11th Annual Research Festival*, ECE, USC, 2021.
15. Invited talk of "Provable constrained policy optimization for reinforcement learning" in *the 38th Southern California Control Workshop*, University of California, Irvine, California, Virtual, 2021.
16. Invited poster of "Distributed robust statistical learning: Byzantine mirror descent" in *the 10th Annual Research Festival*, ECE, USC, 2019.
17. Invited poster of "An exponentially stable primal-dual algorithm for nonsmooth optimization" in *the 9th Annual Research Festival*, ECE, USC, 2018.
18. Invited talk of "A primal-dual algorithm for distributed resource allocation" in *the 34th Southern California Control Workshop*, University of California, Riverside, California, 2018.

Conference Talks & Posters

1. Contributed poster of "Convergence of natural policy gradient primal-dual methods for constrained convex MDPs" in *the 8th Annual Learning for Dynamics and Control Conference*, Los Angeles, California, 2026.
2. Contributed poster of "Composition and alignment of diffusion models using constrained learning" in *the 39th Conference on Neural Information Processing Systems*, San Diego, California, 2025.
3. Contributed poster of "Alignment of large language models with constrained learning" in *the 39th Conference on Neural Information Processing Systems*, San Diego, California, 2025.
4. Contributed poster of "Deterministic policy gradient primal-dual methods for continuous-space constrained MDPs" in *the 39th AAAI Conference on Artificial Intelligence*, Philadelphia, Pennsylvania, 2025.
5. Contributed poster of "Constrained diffusion models via dual training" in *the 38th Conference on Neural Information Processing Systems*, Vancouver, Canada, 2024.
6. Contributed poster of "One-shot safety alignment for large language models via optimal dualization." in in *the 38th Conference on Neural Information Processing Systems*, Vancouver, Canada, 2024.

7. Contributed talk & poster of "Last-iterate convergent policy gradient primal-dual methods for constrained MDPs" in *the 37th Conference on Neural Information Processing Systems*, New Orleans, Louisiana, 2023.
8. Contributed poster of "Sample efficient generalized Lagrangian policy optimization for safe MARL" in *the 5th Annual Learning for Dynamics and Control Conference*, Philadelphia, Pennsylvania, 2023.
9. Contributed talk of "Policy gradient primal-dual mirror descent for constrained MDPs with large state spaces," in *the 61st IEEE Conference on Decision and Control*, Cancún, Mexico, 2022.
10. Contributed talk of "Policy gradient primal-dual method for constrained MDPs," in *the 2022 American Control Conference*, Atlanta, Georgia, 2022.
11. Contributed talk & poster of "Independent policy gradient for large-scale Markov potential games: sharper rates, function approximation, and game-agnostic convergence" in *the 39th International Conference on Machine Learning*, Baltimore, Maryland, 2022. (**118/1117 long presentations**)
12. Contributed talk & poster of "Provably efficient safe exploration via primal-dual policy optimization" in *the 24th International Conference on Artificial Intelligence and Statistics*, Virtual, 2021. (**48/455 orals**)
13. Contributed talk & poster of "Natural Policy Gradient Primal-Dual Method for Constrained Markov Decision Processes" in *the 34th Conference on Neural Information Processing Systems*, Virtual, 2020.
14. Contributed talk of "Global exponential stability of primal-dual gradient flow dynamics based on the proximal augmented Lagrangian" in *the 59th IEEE Conference on Decision and Control*, Virtual, 2020.
15. Contributed poster of "Fast multi-agent temporal-difference learning via homotopy stochastic primal-dual method," in *the Optimization Foundations for Reinforcement Learning Workshop at NeurIPS*, Vancouver, Canada, 2019; *the Southern California Machine Learning Symposium*, UCSD, 2020.
16. Contributed talk of "Exponential stability of primal-dual gradient flow dynamics based on proximal augmented Lagrangian," in *the 2019 American Control Conference*, Philadelphia, Pennsylvania, 2019.
17. Contributed talk of "Nonsmooth composite minimization: an exponentially convergent primal-dual algorithm," in *the 57th IEEE Conference on Decision and Control*, Miami Beach, Florida, 2018.
18. Contributed talk of "A primal-dual Laplacian gradient flow dynamics for distributed resource allocation problems," in *the 2018 American Control Conference*, Milwaukee, Wisconsin, 2018.
19. Contributed talk of "Adaptive Mittag-Leffler stabilization of commensurate fractional-order nonlinear systems" in *the 53rd IEEE Conference on Decision and Control*, Los Angeles, California, 2014.

20. Contributed talk of "Alternative LMI characterizations for fractional-order linear systems" in *the 33rd Chinese Control Conference*, Nanjing, China, 2014.

21. Contributed talk of "Fractional-order integral state space modeling" in *the 26th Chinese Control and Decision Conference*, Changsha, China, 2014.

HONORS & AWARDS

Generative Artificial Intelligence (GenAI) Faculty Fellowship, The University of Tennessee, Knoxville 2026

Sigma Xi Membership, Sigma Xi, The Scientific Research Honor Society 2026

Scholar Award, Conference on Neural Information Processing Systems 2023

Expert Reviewers, International Conference on Machine Learning 2021

Travel Award, Conference on Neural Information Processing Systems 2020

Top Reviewers, International Conference on Machine Learning 2020

Travel Award, IEEE Conference on Decision and Control 2020

Travel Award, American Control Conference 2018, 2019, 2022

MHI PhD Scholar Finalist, ECE, University of Southern California 2018, 2021

ECE Department Fellowship, University of Minnesota 2015

Honor for Outstanding Graduate Student, Zhejiang University 2015

Nominate Paper Award, LSMS & ICSEE, 2014, Shanghai 2014

Bosch Scholarship, Bosch in China 2013

The First-Class of Graduate Scholarship, Zhejiang University 2012–2015

National Scholarship, Ministry of Education of P.R. China 2011

Wei Shaoxiang Engineering Talent, Wei Shaoxiang Foundation, HongKong 2010

The Second-Class of Physics and Technology Innovation Contest, Zhejiang Physical Society, Zhejiang, China 2009

The First-Class of Advanced Mathematics Contest, Zhejiang Mathematical Society, Zhejiang, China 2008

ACADEMIC SERVICE

Journal Referee

IEEE Transactions on Control of Network Systems
IEEE Transactions on Automatic Control
IEEE Robotics and Automation Letters
IEEE Transactions on Intelligent Vehicles
IEEE Control Systems Letters
Systems & Control Letters
Optimization Letters
Machine Learning
Neural Networks
Automatica
IEEE Transactions on Pattern Analysis and Machine Intelligence
APSIPA Transactions on Signal and Information Processing
Engineering Applications of Artificial Intelligence
SIAM Journal on Mathematics of Data Science
Transactions on Machine Learning Research

Journal of Machine Learning Research
IET Control Theory & Applications
International Journal of Robust and Nonlinear Control
Frontiers of Information Technology & Electronic Engineering
International Journal of Systems Science
The Journal of the Franklin Institute
Journal of Applied Mathematics and Computing
Nonlinear Dynamics
IEEE Access

Conference Referee

IEEE Conference on Decision and Control 2018–2026
Conference on Neural Information Processing Systems 2020–2026
International Conference on Learning Representations 2021–2027
International Conference on Artificial Intelligence and Statistics 2021, 2025, 2026
International Conference on Machine Learning 2020–2026
ICML Workshop on Next Generation of AI Safety 2024
ICML Workshop on Theoretical Foundations of Foundation Models 2024
NeurIPS Safe Generative AI Workshop 2025
Learning for Dynamics and Control Conference 2024, 2025
AAAI Conference on Artificial Intelligence 2023, 2024, 2026, 2027
ACM Urban-AI Workshop 2026
American Control Conference 2018–2025
IFAC World Congress 2020
Chinese Control Conference 2014
Chinese Control and Decision Conference 2014

Area Chair

International Conference on Learning Representations, San Francisco 2027
NeurIPS Workshop on Constrained Optimization for Machine Learning, San Diego 2025

Section Chair for Advances in Safe, Robust, and Constrained Decision-Making

65th IEEE Conference on Decision and Control, Honolulu, Hawaii 2026

Section Co-Chair for Nonlinear System and Control

26th Chinese Control and Decision Conference, Changsha, China 2014

Admissions Committee, University of Pennsylvania

PhD Admission in the Department of Electrical and Systems Engineering Fall 2023

Conference Volunteer

38th International Conference on Machine Learning, Virtual 2021, 2022
24th International Conference on Artificial Intelligence and Statistics, Virtual 2021
35th Conference on Neural Information Processing Systems, Virtual 2021

TEACHING & CURRICULUM DEVELOPMENT

Instructor, University of Tennessee, Knoxville
COSC 594 *Interplay between Reinforcement Learning and Generative Artificial Intelli-*

gence Fall 2025
ECE 315 *Signals and Systems* Spring 2026
COSC 594/650 *Optimization for Machine Learning* Fall 2026

Lab Assistant, University of Pennsylvania
ESE 2000 *Artificial Intelligence Lab: Data, Systems, and Decisions* Summer 2024

Guest Lecturer, University of Pennsylvania
ESE 5140 *Graph Neural Networks* Fall 2023

Teaching Assistant, University of Southern California
EE 587 *Nonlinear Systems* Spring 2018

Guest Lecturer, University of Minnesota Twin Cities
EE 3015 *Statistical Methods in Electrical and Computer Engineering* Spring 2017

Teaching Assistant, University of Minnesota Twin Cities
EE 4231 *Automatic Control Systems* Fall 2016
EE 3015 *Statistical Methods in Electrical and Computer Engineering* Spring 2017
EE 8231 *Optimization Theory* Spring 2017

ADVISING & MENTORING

Doctoral Student Advisee, University of Tennessee, Knoxville
Anik Saha, Electrical Engineering and Computer Science August 2026 – Now
Topic: *Constraint-led generative modeling*

Master Student Mentee, University of Tennessee, Knoxville
Jisu Park, Electrical Engineering and Computer Science June 2026 – Now
Topic: *Constrained reinforcement learning*

Undergraduate Student Mentee, University of Tennessee, Knoxville
Mitchell Loh, Haslam College of Business March 2026 – Now
Topic: *Constrained game theory*

PhD Committee Member, University of Tennessee, Knoxville
Fanqi Wang, Electrical Engineering and Computer Science December 2025

Doctoral Student Mentees, University of Pennsylvania
Shervin Khalafi (4th year) September 2023 – Now
Topic: *Constrained learning for diffusion models*

Ignacio Hounie September 2024 – Fall 2025
PhD Thesis: *Learning with pointwise constraints*
Now: *Senior Researcher, Microsoft*

Xinmeng Huang February 2024 – February 2025
PhD Thesis: *Topics in statistical machine learning: multitask learning, uncertainty quantification, and language model alignment*
Now: *Quantitative Researcher, Citadel Securities*

Doctoral Student Mentee, King Juan Carlos University, Madrid, Spain
Sergio Rozada February 2024 – February 2025
PhD Thesis: *Low-rank methods in reinforcement learning*

Now: *Assistant Professor, King Juan Carlos University (Spain)*

Master Student Mentee, University of Pennsylvania
Zhengyan Huan — September 2023 – February 2024
Topic: Resilient constrained reinforcement learning
Now: *Doctoral Student, Tufts University*

Undergraduate Student Mentee, University of Southern California
Terry Lu — January 2022 – June 2022
Now: *Trading Analyst, Symmetry Investments*

Academic Mentor for Master Students, University of Southern California
Viterbi Graduate Mentorship Program — August 2018 – June 2021

SKILLS

Python, LaTeX, C/C++, Matlab, SQL

MEMBERSHIPS

The Institute of Electrical and Electronics Engineers (IEEE)
IEEE Control Systems Society Membership
The Institute for Operations Research and the Management Sciences (INFORMS)
Sigma Xi
American Association for the Advancement of Science (AAAS)
Stanford Encyclopedia of Philosophy

GRADUATE COURSE HIGHLIGHTS

Control Systems: EE 5231 *Linear System and Optimal Control*, EE 8215 *Nonlinear Systems*, AEM 8421 *Robust Multi-Variable Control Systems*, AEM 8423 *Convex Optimization Methods in Control*; **Optimization and Computation:** EE 5239 *Introduction to Nonlinear Optimization*, EE 8231 *Optimization Theory*, ISE 633 *Large-Scale Optimization for Machine Learning*, CSCI 5304 *Computational Aspects of Matrix Theory*, CSCI 8314 *Sparse Matrix Computations*; **Probability and Machine Learning:** MATH 507A/B *Theory of Probability*, EE 556 *Stochastic Systems & Reinforcement Learning*, CSCI 5525 *Machine Learning*, EE 546 *Mathematics of High-Dimensional Data*, DSO 699 *Statistical Learning Theory*, CSCI 699 *Theoretical Machine Learning*, EE 5581 *Information Theory and Coding*.